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“Ethical Responsibility Very Often Gets Drowned Out”: A Qualitative Interview Study of Genome Scientists’ and ELSI Scholars’ Perspectives on the Role and Relevance of ELSI Expertise

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ABSTRACT

Background: Genome scientists and Ethical, Legal, and Social Implications of genetics (ELSI) scholars commonly inhabit distinct research cultures – utilizing different research methods, asking different research questions, and valuing different types of knowledge. Collaborations between these two communities are frequently called for to enhance the ethical conduct of genomics research. Yet, little has been done to qualitatively compare genome scientists’ and ELSI scholars’ perspectives on collaborations with each other and the factors that may affect these collaborations.

Methods: 20 semi-structured interviews with US-based genome scientists and ELSI scholars were conducted between June–September 2021. Interviews were analyzed using inductive thematic analysis.

Results: Genome scientists and ELSI scholars provided different understandings of the value and goals of their collaborations with each other. Genome scientists largely perceived ELSI expertise to be relevant for human subjects research; they described ELSI scholars as communicators who help the public and/or study participants better understand genomics research. In comparison, ELSI scholars viewed themselves as developing and implementing policies; they expressed frustration at how scientists can misunderstand their research methods or negatively perceive them. A combination of factors – both structural (e.g., criteria for promotion) and cultural (e.g., perceptions of what colleagues value and respect) – seemed to shape these diverging perspectives.

Conclusion: Academic institutions, funders, and researchers commonly call for collaborations between genome scientists and ELSI scholars, but under-consider how their different conceptual frameworks, research methods, goals, norms, and values, conjoin to affect such partnerships. Acknowledging, exploring, and addressing the complex interplay between these factors could help to more effectively facilitate collaborations between genome scientists and ELSI scholars.

KEYWORDS

Research cultures; genomics; ELSI; research collaboration

Introduction

The completion of the Human Genome Project in 2003 fueled significant advancements in our understanding of the contribution of genetics to human health and well-being. It also reinforced the need to consider the ethical, legal, and social implications (ELSI) of genetic/genomic data, research, and applications. Historically, claims regarding genetic differences in human health and behavior were wielded in service of socially harmful rhetoric and action. Today, these dangerous mischaracterizations continue, and

there is concern that ongoing genomics research, especially on social and behavioral traits, may be used to bolster them. For example, the White teenager who murdered 10 African Americans in a supermarket in Buffalo, New York in May of 2022 cited in his screed several genomic studies as evidence of biological differences between racial groups and the genetic superiority of White people over Black people in terms of intelligence (Wedow, Martschenko, and Trejo 2022).

A frequent recommendation for how to appropriately identify and safeguard against the misuse and misinterpretation of genomics research has been to

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encourage collaborations between those who generate and utilize genomic data (i.e., genome scientists) and those who investigate the social and ethical implications of such data (i.e., Ethical, Legal, and Social Implications, or ‘ELSI,’ scholars) (Martschenko and Trejo 2022; National Academies of Sciences, Engineering, and Medicine 2023; Reardon et al. 2023). Since the 1980s, U.S. federal resources and funding have been allocated to scholarly investigations of the ELSI of Human Genetics (Juengst 2021). Funders and research institutions continue to emphasize the importance of collaborations between researchers in Human Genetics and ELSI scholars in strategic plans (Green et al. 2020) and funding announcements. For instance, at the behest of federal funders, ELSI scholars in the United States are often embedded in large life science projects e.g., (Dolan, Lee, and Cho 2022), “as a prescribed part of human genome research.” (Juengst 2021)

Importantly, however, genome scientists and ELSI scholars tend to inhabit different research cultures. That is, they often propose, communicate, evaluate, legitimize, and generate knowledge using different conceptual frameworks, methods, tools, and practices (Cetina 1999; Nelson and Panofsky 2018; Yuan 2022). For instance, genome scientists are more likely to hold a “technological orientation” (Kastenhofer 2007); they primarily use quantitative research methods to answer empirical research questions about the structure, function, mapping, and evolution of the genome. Genome-wide association studies (GWAS) enable genome scientists to conduct large-scale investigations of whether and how genetic differences between individuals relate to differences in health and well-being (Bush and Moore 2012). And, many genome scientists are working to develop predictive tools, like polygenic scores (Centers for Disease Control and Prevention 2022), that estimate a person’s likelihood of developing a disease or outcome.

In comparison, ELSI scholars tend to focus on developing “precautionary practices.” (Kastenhofer 2007) That is, ELSI scholars primarily strive to anticipate and respond to emerging scientific issues in fields like Human Genetics “before science provokes them as practical policy issues.” (Dolan, Lee, and Cho 2022) They ask empirical as well as normative research questions, and investigate the ethical, legal, and social challenges raised by genomics research using a variety of research methods, including experimental survey vignettes, interviews, and deliberative democratic engagements (Parker et al. 2019). In short, ELSI scholars and genome scientists frequently adopt different approaches to producing knowledge, rational analysis, and empirical inquiry, and subsequently generate different kinds of knowledge (Kastenhofer 2007; Yuan 2022).

Despite moves from funders and research institutions to foster collaborations between ELSI scholars and genome scientists, and despite a growing public conversation about the ethical and social implications of genomics research, little has been done to qualitatively explore genome scientists’ and ELSI scholars’ perspectives on their collaborations with each other. Our study sets out to fill this gap. We draw on a larger qualitative interview study on the barriers, facilitators, harms, and benefits of interdisciplinary collaboration to explore genome scientists’ and ELSI scholars’ views on both the relevance of collaborations with each other and their roles in these collaborations. To collaborate ELSI scholars and genome scientists will need to find at least some degree of convergence between the different conceptual frameworks, research questions, and methods of each. Thus, it is important to understand how these groups currently converge and/or diverge in their thinking about when and how to collaborate with each other.

Our findings highlight a complex interplay between factors like the tenure and promotion process and differences in terms of what kind of knowledge is considered valuable. In interviews, genome scientists held relatively narrow conceptualizations of ELSI scholars – describing them as relevant for genomics research that directly involves human subjects and useful for aiding with the communication of genomics research findings, such as during the return of results to patients. ELSI scholars, in comparison, described themselves as developing policies and implementing them in-tandem with diverse stakeholder groups. Many also described feeling misunderstood or disregarded by scientists who view ELSI scholars as burdens or irrelevant.

We argue that if facilitating “robustly interdisciplinary initiatives” (Juengst 2021) between ELSI scholars and genome scientists is considered instrumental to the responsible conduct and translation of genomics research, then it will be necessary to identify and understand the factors that affect these collaborations and how they inform each other. More thoroughly understanding how genome scientists and ELSI scholars think about their collaborations with each other, and how various factors may conjoin to affect these collaborations, could beneficially illuminate frictions that stand in the way of success.

Methods

Participant recruitment

This study includes 20 in-depth, semi-structured interviews conducted between June and September 2021 with US-based genome scientists and ELSI scholars.

We specifically sought to recruit genome scientists working in or adjacent to social and behavioral genomics and ELSI scholars focused on human genomics research. There were two primary reasons for this focus. First, social and behavioral genomics – a field that investigates whether and how genetic differences between individuals affect differences in behaviors and social outcomes (Harden and Koellinger 2020) – is the subject of growing ELSI interest and concern (Martschenko 2021; Martschenko, Trejo, and Domingue 2019; Matthews et al. 2021; Sabatello, Martin, et al. 2021; Sabatello, Insel et al. 2021). The National Human Genome Research Institute's most recent strategic vision, for instance, explicitly calls for scholarly investigations of “the implications of studying genetic associations with bio-behavioural traits (such as intelligence, sexual behaviour, social status, and educational attainment).” (Green et al. 2020) Second, we wanted to gather the perspectives of individuals who might conceivably collaborate with each other, or benefit from doing so. Recent research efforts have intentionally sought to bring together genome scientists and ELSI scholars to examine the risks, benefits, and ethical responsibilities of social and behavioral genomics – demonstrating an existing belief that such collaborations may be beneficial (Meyer et al. 2023).

Participants were randomly selected from one of two purposive datasets generated by the research team. The first dataset ($n=1163$) includes a list of all authors with publications between 2016 and 2021 in one of the following top-tier academic journals where social and behavioral genomics research is published: a *Nature*- or *Science*-affiliated journal, *Behavior Genetics*, or *Cell*. Publications needed to include the keywords ‘GWAS’ and/or ‘genome-wide association study’ and ‘educational attainment’ and/or ‘intelligence.’ These selection criteria sought to recruit individuals who conduct GWAS on, or related to, social/behavioral phenotypes like educational attainment and intelligence – traits that evoke staunch academic debate. As such, our study specifically solicited the perspectives of genome scientists who focus on genetic and mechanistic discovery and methods development rather than those who may more frequently interact with patients and study participants (e.g., genome scientists in medical genetics, clinical genetics, and/or genetic epidemiology).

The second dataset ($n=206$) includes a list of all authors with publications between 2016 and 2021 in one of the following highly-ranked academic journals where ELSI research is published: *Genetics and Medicine*, *Journal of Law, Medicine, and Ethics*, the

American Journal of Bioethics, the *Journal of Empirical Research on Human Research Ethics*, or *Public Health Genomics*. Publications needed to include the keywords ‘ELSI’ and ‘bioethics.’¹ Publications also needed to contain the keyword ‘genetics’ and/or ‘genomics.’ These selection criteria sought to enrich for individuals who study ELSI questions related to genetics/genomics research.

Ten individuals from each database were randomly selected and screened to ensure they were US-based. In total, 17 individuals were contacted using the first database. Of those contacted, six did not respond. Of the 11 individuals that responded, all were interviewed.² Twelve individuals were contacted using the second database. One did not respond, and one declined to participate.³ The remaining 10 individuals were interviewed.

Data collection

After random selection and checks to ensure fit with screening criteria, individuals were emailed an invitation to take part in an hour-long confidential Zoom interview about interdisciplinary collaboration and its barriers and facilitators. Prior to interview, all participants filled out a confidential questionnaire to collect basic demographic information and provide informed consent. Interviews began by asking participants about their research interests and the policy implications, if any, of their work. All participants were asked to describe what interdisciplinary collaboration meant to them and its purpose. The term ‘interdisciplinary collaboration,’ was purposefully left undefined and open-ended by the interviewer (DOM). Participants were also asked to describe collaborations they had been a part of that they considered to be successful/unsuccessful and to reflect on what had made the collaborations successful/unsuccessful. This was meant to illuminate potential barriers and facilitators. Finally, participants shared their lived experiences specifically collaborating, or not, with an ELSI scholar or genome scientist. In doing so, they were asked to describe the context that led or did not lead to a collaboration. This paper examines this latter line of questioning. Upon conclusion of interview, participants received a \$100 Amazon gift-card.

Data analysis

Interviews were audio recorded, transcribed by a third-party service, and checked for quality.⁴ All personal identifiers were removed, and anonymized transcripts were entered into NVivo qualitative data

analysis software. DOM employed thematic analysis on all interviews to identify initial patterns, or ‘themes’ across a data set (Clarke and Braun 2014) and developed a preliminary codebook. Then DOM and AG met weekly and double-coded ten randomly selected interviews to refine the codebook and ensure congruence in analysis. MKC was consulted when coding incongruencies between DOM and AG could not be resolved. Once the codebook was finalized, DOM and AG then re-analyzed every third interview to ensure that the finalized codebook sufficiently captured emergent themes.⁵ An inductive approach was taken in thematic analysis (Clarke and Braun 2014).

Results

Participant characteristics

Table 1 provides demographic characteristics of study participants. Over half of interview participants (65%) self-identify as current faculty members and most self-identify as white.⁶ the purpose of qualitative data is not to produce generalizable knowledge. Nevertheless, the skew toward male-identifying faculty is reflective of current postsecondary faculty demographics (National Center for Education Statistics, 2022), especially in STEM. (Kozlowski et al. 2022) Two participants were retired at the time of interview – one was selected *via* the ELSI database and the other *via* the genetics database. Participants hold degrees (primarily PhDs) in an array of disciplines including medicine, philosophy, biochemistry, genetics, anthropology, and the Law. Since the sampling

strategy purposively sought to include social and behavioral genomics researchers, a number of participants who were selected *via* the genome scientist database hold degrees in social sciences disciplines such as economics, which is common for the field. (Bliss 2018).

Themes

In this section we present the main themes to emerge from inductive thematic analysis.

Genome scientists’ perspectives on the relevance of ELSI expertise

Three of the ten genome scientists (2-G, 6-G, and 12-G) stated that they had directly collaborated with an ELSI scholar; all three are tenured faculty and none mentioned that they had actively sought out ELSI expertise themselves. Instead, they each described a fortuitous moment or opportunity that had led them to engage with an ELSI scholar.

A number of genome scientists explained that they had not collaborated with an ELSI scholar because they do not consider ELSI expertise relevant to their specific kind of research. For example, 17-G explained that he had not collaborated with an ELSI scholar because his research does not involve human subjects. 17-G conducts GWAS, which frequently fall outside the realm of human subjects research. (To run GWAS, researchers tend to pull data from large biobanks.) 17-G described himself as “*more of a basic researcher*” and felt that “*when... we do work sort of maybe more*

Table 1. Participant characteristics.

ID#	Race/ethnicity	Gender	Position	Training discipline	Current discipline
Genome Scientists					
2-G ^a	White	Man	Faculty	Psychology (PhD)	Genetics/Neuroscience
5-G	Middle Eastern	Woman	Postdoc	Psychology (PhD)	Psychology
6-G	White	Man	Faculty	Economics (PhD)	Economics
7-G	Asian	Woman	Faculty	Genetics (PhD)	Genetics
9-G	White	Man	Faculty	Economics (PhD)	Economics, Sociology
12-G	White	Man	Faculty	Biochemistry (PhD)	Genetics
15-G	White	Man	Retired Faculty	Cell Biology and Anatomy (PhD)	Neuroscience (Now Retired)
16-G	Asian	Man	Research Staff	Physics (PhD)	Computational Biology and Bioinformatics
17-G	Middle Eastern	Man	Faculty	Genetics (PhD)	Human Genetics
18-G	White	Woman	Postdoc fellow	Psychology - Behavioral Genetics (PhD)	Psychology - Behavioral Genetics
ELSI Scholars					
1-E	White	Man	Faculty	Medicine (MD)	Bioethics
3-E	Asian	Woman	Faculty	Medicine (MD)	Medicine (Pediatrics)
4-E	Hispanic	Woman	Faculty	Philosophy (PhD)	Bioethics, Philosophy
8-E	Asian	Woman	Research Staff	Medical anthropology (MA)	Bioethics
10-E	American Indian	Woman	Faculty	Public Health (PhD)	Intellectual and Developmental Disabilities
11-E	White	Woman	Retired	Social Policy (PhD)	ELSI
13-E	White	Man	Faculty	Medicine (MD)	Public Policy
14-E	White	Woman	Faculty	Education Chemistry and Math (PhD)	Evaluation
19-E	White	Woman	Faculty	Anthropology (PhD)	ELSI
20-E	White	Woman	Attorney	Law (JD)	Law

^aG’ signifies that interviewee is a genome scientist and ‘E’ signifies that the interviewee is an ELSI scholar.

closely with impact [on] individuals, we then have to start thinking about ethics.” He later shared: “my work is not really... we’ve not done anything directly with returning information to patients. If I did, I would definitely involve someone from there [bioethics institute] on my grant proposal.” Thus, although 17-G considers ELSI expertise relevant for certain kinds of genomics research, he does not consider it of value to his own scholarship.

Similarly, 7-G described her work as focusing “on finding biomarkers or finding genetic regions.” She explained:

I think I’ve never had a chance to think beyond my findings, like for example I also participate in genome-wide association studies, and we identified polygenic risk scores in those, but I think somehow I thought that my job is to identify those and then after that—how those could be applied to the real clinics... I think I haven’t had a chance to think about those [sic].

She went on to say:

For me as a junior faculty I’ve been more focused on the core, which is the molecular, you know, biology/genetics and the research, so I think that was why so far I haven’t had a chance to, yeah, work with others like you [referring to DOM].

Genome scientists cited additional factors that they felt explained why they had not collaborated with an ELSI scholar. For instance, 7-G also shared that her grants do not require her to consider or incorporate ELSI perspectives:

I think that molecular biology or the people with similar background as me... [have] just not had a chance to work with [ELSI scholars]... but I’m sure that more like epidemiologists or other people ... [would] be happy to, you know, get more involved [with them]... But for me it’s quite – yeah, I need to do the research that I’m funded currently [to do].

Moreover, she described having “never had a chance to meet or know about their [ELSI scholars’] interests” because of academia’s siloed disciplinary structure. As someone who works in a biochemistry/molecular medicine department with colleagues who have “*diplo-mas in biochemistry, biology, or genetics*” and come from “*similar background[s]*,” 7-G did not know of any ELSI scholars – explaining that they “*join different PhD program[s]*.”

Other genome scientists also pointed to other issues that diminished their perceived relevance of ELSI expertise. 16-G, for example, described wanting to collaborate with ELSI scholars because he felt “*there*

needed to be more sensitivity in the way that we discuss [psychiatric genetics findings]” – his area of specialization. He explained that he thought “*bioethicists need to be frequently involved in the kind of research we do.*” However, while he is interested in working with ELSI scholars, he expressed that he feels alone among his colleagues in his desire for such research partnerships:

It would have helped to have a little bit more like okay, there’s a few of us who are interested, you know, gather some kind of community that is interested and pushing that [exploring ELSI issues]. I kind of feel like I suggested it on my own, pursued it on my own, failed on my own, and then as a result kind of it went nowhere.

16-G went on to share that he thought the requirements for securing tenure and promotion (e.g., publication expectations) can encourage genome scientists to deprioritize their ethical responsibilities. He himself is subject to these criteria for promotion as an early career scholar:

The truth is that yes, everyone wants to get their papers published and, you know, we want to move on in our careers, we want to do this for the next stage and next stage, and so... there isn’t currently much of a motivator to – drivers towards reaching out, other than our own sort of feeling of ethical responsibility, which very often gets drowned out unfortunately.

Although the majority of genome scientists interviewed had not worked with an ELSI scholar, a few (9-G, 16-G, 18-G) described unsuccessful attempts to find and collaborate with one. For instance, 9-G offered a couple of reasons for why he had been unsuccessful at securing ELSI expertise. These included the senior career stage of his potential collaborators and his perception that those he approached “*on the ethics side*” had not considered his ideas or request to collaborate worthwhile:

I wouldn’t say I have any projects that I can think of that brought in that [an ELSI] perspective. I mean there’s been several... I’d say there’s a handful of not false starts, but just conversations that I hoped would have led to a collaboration that didn’t and I would chalk up the reasons that it didn’t [proceed] under like generic reasons, which is that the people who I’ve tried mostly are here [home institution] and some of them are senior and I would say like just have their own – they’ve got their own work ... it’s the people who I’ve spoken to more on the ethics side saying, you know, ‘really interesting, these are great, these are things that we should do’ like in general terms, but just not having time to right now and like pushing that aside...

Echoing 9-G, 18-G shared:

I don't think the [ELSI] individuals we've approached have had time to fully collaborate with us, so they've more like 'here, let me do a session with you and tell you about some of the things that we think are important factors'... I know of about two people at our university who have both a kind of sufficient understanding of the type of research we do and are considered a bioethicist, but they don't have the time to dedicate.

In summary, a majority of genome scientists had not collaborated with an ELSI scholar. Several explained that this was because ELSI expertise is relevant for human subjects research, which they do not conduct. Other genome scientists shared that the ELSI scholars they had approached seemed too busy to collaborate with them. Additional barriers to seeking out ELSI expertise identified by genome scientists included their perceptions of the research funding available to support them in collaborating with ELSI scholars, what their genomics colleagues value and respect, and current criteria for promotion.

ELSI scholars' perspectives on the relevance of ELSI expertise

A number of ELSI scholars expressed the view that ELSI expertise is relevant for basic sciences research as well as human subjects research. On the one hand, some ELSI scholars described productively working with basic scientists who valued ELSI expertise and saw it as relevant to their work. For example, 1-E shared:

There's a species of people in basic science who appreciate that there's really interesting [ethical] issues that they're confronting and are trying to figure how do they get more engaged with it.

On the other hand, however, ELSI scholars like 1-E, 3-E, 14-E, and 20-E shared their personal experiences encountering scientists who they felt had failed to appreciate or understand the relevance of ELSI expertise. Several of the genome scientists interviewed for this study considered ethical expertise irrelevant to their basic sciences research; 14-E had experienced this sentiment during her work with bench scientists. She explained:

I do most of my work with scientists, regular bench scientists. So, you know, [what] I will say is [that] particularly engineers are not particularly ethical-interested in ethics as much. You know, comments like 'well I'm just an engineer'...so that whole idea of them being- needing to understand ethical issues as they're creating their devices is kind of new.

In the same vein, 1-E, when describing his exchange with a scientist he had hoped to collaborate with, stated:

I just got a sense he really wasn't interested. And I think it really was not that he was opposed to it, it was just not part of his vision.

And, 3-E explained:

Making people sort of recognize the value in other disciplines... I think it can be difficult to get those people even in the same room and do the scientists want to hear about ethics? I don't know. Maybe they think they're too busy for it.

Like 3-E, other ELSI scholars also described scientists as being "*super busy*" (20-E), or at least considering themselves to be too busy for ELSI collaboration. In this way, several ELSI scholars mirrored the descriptions of genome scientists who described ELSI scholars as being too busy to collaborate with them. For instance, when recounting another unsuccessful collaboration with a scientist, 1-E expressed his frustration:

He wouldn't even meet with me. And I really had a sense that he viewed- he had this impression of what I did as somebody who was just going to slow him down, and no conversation whatsoever, I've never spoken to him ever since... he has zero interest in any sort of 'let's think about the ethical issues of this'.

1-E later described yet another failed collaboration. This time, his potential collaborator had "*had a previous experience with people in bioethics*." However, this previous experience seemed to negatively affect his perception of what 1-E could offer him. 1-E explained:

In his [the collaborator's] mind we [ELSI scholars] were just slowing everything down. And so, he- our relationship was really very unpleasant, and it didn't go very well, and we eventually stopped working together.

As another example, 11-E raised the point that 'hard' scientists can dismiss 'soft' scientists as being less serious or knowledgeable. Furthermore, she explained that there seems to be a power differential between these two research communities:

A lot of hard science people have stereotypes about the soft social sciences, 'cause they think it's not really serious or they think they [social scientists] don't really understand the science and so anything that they say is kind... you know, they just dismiss it and plus there's the whole power differential thing, you know, the... there's always been a sense that people in the hard sciences, have greater power than those in the social sciences

Finally, 8-E who is Research Staff and trained in qualitative methods shared her experiences working with quantitative-oriented collaborators who were scientists and medical professionals. She explained: “*other researchers who are not qualitative researchers have not been excited about those [qualitative] methods. They don’t really like that approach, they don’t really because they want to be more quantitative.*”

In short, ELSI scholars considered their expertise widely applicable. For them, ethical issues emerge in relation to all kinds of scientific research – regardless of whether or not human subjects are involved. Nevertheless, a few discussed the struggle they sometimes face to convince scientists of the relevance of ethics and ELSI expertise.

Genome scientists’ perspectives on the role of ELSI scholars

When asked to describe when they had or might consider collaborating with an ELSI scholar, more than half of genome scientists (5-G, 7-G, 15-G, 16-G, 17-G, 18-G) shared that ELSI scholars could support them with research communication. 18-G, for instance, expressed:

We have brought in [ELSI] individuals into our lab to speak with us about conducting research ethically I guess, but we haven’t necessarily collaborated with them on a project... I think I would definitely be very interested in trying to particularly bring in, you know, somebody for the – particularly the conclusions, discussion, implications sections, ‘cause I think that’s one of the areas where it could really benefit... and then like if we could do, I don’t know, consulting sessions, be like here’s our specific issue, like what would your approach to this be.

Along similar lines, 15-G explained:

I guess in the end my answer is no we never really sort of incorporated an ethics person. I can see how that might have been advantageous to at least how we might go about shaping conversations in one way or another.

15-G also provided a specific example of a project that, in retrospect, he thought could have benefited from collaboration with an ELSI scholar. His research had received pushback from some members of the public, and he felt that an ELSI scholar could have helped him more effectively communicate with those who were resistant to it:

I think having someone to help me and help them walk through the ethics of conducting some of that kind of research might have been very helpful. It might have made my life a lot less stressful!

5-G, had “*been thinking about particularly the communication aspect for a pretty long time.*” However, although she felt that a collaboration involving an ELSI scholar could help her think about “*the way that it [behavior genetic research] is perceived, the way that it is being used by people with very sinister motives,*” and assist with “*framing the message,*” she had not engaged ELSI expertise. As an early career scholar who was newly interested in behavior genetic research and was constrained by prevailing expectations for tenure and promotion, she said: “*I feel like I don’t have the... I’m not qualified enough or in the field enough, a leader in the field, that I would be capable to pull that [kind of collaboration] off.*”

12-G who had collaborated with an ELSI scholar to develop a course for geneticists and lawyers stated that the experience helped pose questions “*I’ve never thought about*”; he had found it to be “*very kind of eye-opening.*” However, 12-G also explained that differences exist between genome scientists and ELSI scholars in terms of how each is expected to frame research and the knowledge it generates:

It’s hard for me to publish a paper in science in what I do in a research lab and say we got this result and maybe it means this, maybe it doesn’t. I’m not gonna get away with that. I got to be a little more definitive. In ethics you have to recognize that there are sides to different issues, that there may not be a right answer, but it’s personal decisions and personal impressions that are the important component of that.

In conclusion, a number of genome scientists considered collaborations with ELSI useful for research communication. More specifically, several described ELSI scholars as ‘communicators’ who could help convey the purpose or findings of their scientific research to members of the public or study participants.

ELSI scholars’ perspectives on the role of ELSI scholars

ELSI scholars frequently described themselves as implementors and policy developers. For example, 13-E explained that “*the whole idea of ELSI was supposed to be let’s do research that informs policy.*” He felt that ELSI scholarship ought to be “*very real world oriented,*” and raise questions that have “*to do with law and policy and legislation that would need to be formulated and created.*” Along similar lines, 20-E stated that one of the goals of ELSI work is to figure out “*how can we implement genetics into clinical care.*” And, 10-E described a specific project in which she was leading focus group discussions with tribal

communities that “*might inform policy procedure...and inform leaders’ decision-making.*”

However, in order to do the policy and implementation work that they described, several ELSI scholars identified scientists as critical partners. In fact, ELSI scholars often described themselves working with an array of different stakeholders, including – as 8-E put it– people “*with law degrees, genetics backgrounds or science backgrounds, or medical backgrounds.*” In comparison to the genome scientists who were interviewed – the majority of whom had not collaborated with an ELSI scholar – all ELSI scholars described collaborating with a scientist (though not necessarily a genome scientist). 11-E, for instance, shared: “*I was always working with geneticists, ‘cause what did I know about genetics, at least at first.*” 20-E also considered scientists necessary collaborators, explaining:

I kind of want to hear from the scientists and physicians what they think the quote “right” answer is. Sort of if you could wave a magic wand, what would you think was best for patients, and then starting from there we can think about how would you get there from a policy perspective? That perspective... it can be hard to pin that down and it can be challenging. When you don’t have the right technical subject matter experts in the room, you wind up doing a lot of speculating among people...

Despite viewing scientists as important contributors to their work, a number of ELSI scholars voiced their dismay at how little scientists seemed to feel the reverse was true: that ELSI scholars are important contributors to scientists’ work. Some ELSI scholars shared that meaningful integration between ELSI scholars and scientists is limited. A few explained that ELSI perspectives are often brought in as an afterthought. 13-E, for example, shared:

The interdisciplinary stuff within an academic institution usually means yeah, I’m doing biology, but then I’m gonna have a chemist and a physicist working with me and oh my gosh, yeah, maybe we’ll think about the ELSI thing, we’ll throw a chapter at the end of our book and we’ll... you know, we’ll hire [name omitted] to write that for us or something, right. And that’s very much the way... and it hasn’t changed all that much.

Some ELSI scholars also discussed the role of ELSI in research communication (e.g., 8-E: “*I think having more voices to help develop that communication and develop fostering a broader discussion. is the point of a lot of bioethics and ELSI research*”). However, they generally did not define themselves as research communicators who help scientists converse with wider audiences – a divergence from the views articulated

by a number of genome scientists who had described ELSI scholars as ‘communicators.’ In closing, ELSI scholars’ primarily described themselves as shaping and implementing policies and decision-making about research and its translation through collaboration with others.

Discussion

Our findings highlight a range of factors – both structural and cultural – that may affect collaborations, between ELSI scholars and genome scientists. We found that a number of genome scientists perceive ELSI to have limited relevance for genomics research, supporting Wolpe’s (2006) observations of how scientists think about ethics (Wolpe 2006) and prior literature on the limited perception of the role of ELSI in life science (Ladd et al. 2009; McCormick, Boyce, and Cho 2009; McCormick et al. 2012). ELSI scholars and genome scientists also shared their perspectives on how promotion criteria and funding availability hinder their collaborations with each other – echoing existing literature on the structural barriers faced by individuals and teams in their pursuit of interdisciplinary and inter-cultural research collaborations (Huutoniemi et al. 2010).

However, our findings also suggest that the factors raised by participants do not operate separately from one another. Instead, there is a complex interplay between them: value-laden expectations about what, how, and with whom to research and publish influence promotion criteria and funding calls, and vice versa. In this study, ELSI scholars and genome scientists articulated different values, norms, and goals for collaborations with each other. At a fundamental level, these differences speak to what each group believes ‘counts’ as knowledge and how they think it should be generated. ELSI scholars and genome scientists may struggle to collaborate because the promotion criteria, conceptual frameworks, research questions, and research methods that each employ are rooted in different values, norms, and goals. For example, both groups described their counterparts as busy and unavailable for collaboration. On the one hand, ELSI scholars and genome scientists may truly have insufficient time available to collaborate with each other. That is, in the rush to write manuscripts, teach, and submit grants, there are only so many hours in the day. On the other hand, time is a limited resource in need of appropriate allocation, and resource allocation decisions entail value-based tradeoffs (Dawson et al. 2020). When determining how to spend time, one must weigh the risks,

potential benefits, cost, materials, and labor that come with any decision.

It is therefore possible, in the context of this study, that ELSI scholars and genome scientists are making value-based tradeoffs when deciding whether or not to collaborate with each other. Genome scientists may *perceive* themselves to be too busy to work with ELSI scholars because they view the type of knowledge that collaborations with ELSI scholars generate to be less valuable to them, less respected by colleagues, or even detrimental to their careers. In comparison, ELSI scholars may *perceive* themselves to be too busy to work with genome scientists because they consider requests to assist scientists with research communication secondary to their goal of developing and implementing policy, or even outside the scope of their work altogether.

Drawing on the scholarship of philosopher Karen Kastenhofer, collaborations between genome scientists and ELSI scholars that position either party as an “ancillary science” can produce an asymmetry that hinders cooperation (Kastenhofer 2007). To promote symmetric cooperation, and by extension the convergence of research cultures, all collaborators should be expected to “take part in the epistemic process itself.” (Kastenhofer 2007) However, several of the genome scientists included in this study seemed to position ELSI scholars in ancillary roles – believing that the value of ELSI expertise lies in assisting scientists with communicating research findings. In other words, ELSI scholars were believed to be necessary for carrying out “standard techniques that their colleagues are not trained or equipped to perform” (in this case talking with patients or members of the public), but neither ELSI scholars nor genome scientists were expected to acquire “knowledge, techniques and material stemming from other scientific fields.” (Kastenhofer 2007) Indeed, elsewhere in the literature ELSI scholars have reflected on their experiences as “tolerated guests” in embedded ELSI projects – expected to perform specific roles that they perceived as limited (Conley et al. 2020).

When interviewees, like 9-G and 18-G, recounted instances where ELSI scholars declined their requests for collaboration, it is worth considering whether one reason for this is because such requests ask ELSI scholars to support scientists with research that they had little to no role in generating. ELSI scholars largely seemed to describe themselves as implementors and policy developers. An invitation to collaborate on research communication efforts may therefore be less important to ELSI scholars who hold different views on the value, relevance, and role of their expertise,

and who are driven by their own promotion criteria. In short, there may be incommensurable differences between ELSI scholars and genome scientists in terms of their perceptions of the role and relevance of ELSI expertise that will need to be addressed if both parties are to find equal or shared value in collaborating with each other (Heidler 2017).

Importantly, it seems unlikely that singularly addressing any one of the factors raised in this study will substantially improve the ease with which these research partnerships are initiated or executed. As Ku and Zehr (2022) point out, even when spatial and human resources are put toward eliminating common structural barriers to research collaborations, challenges to integrating ELSI perspectives into scientific research remain (Ku and Zehr 2022). It is one thing to provide infrastructural support, it is another to navigate different epistemic claims that can create competing research priorities among scientific communities (Ku and Zehr 2022). Institutional obstacles are just one dimension to what can make these collaborations difficult (MacLeod 2018).

This qualitative study is confined to twenty participants at US-based institutions. As is the case with qualitative research, this study is not intended to make generalizable claims about how those working in genomics or ELSI think about their collaborations with each other. However, this study begins to map the factors that may prevent the initiation and successful execution of collaborations involving different research cultures like those of genome scientists and ELSI scholars; it also explores the ways in which these factors may interact with each other. Future research should examine global differences between researchers’ understandings of ethics-genetics collaborations as ELSI is a US-specific concept; common approaches to bringing scientists and ethicists together in the United States (e.g., embedded ethics (McLennan et al. 2020)) may be less common in other contexts that more frequently adopt other conceptual frameworks (e.g., responsible innovation (Balmer et al. 2016)). Thus, it is possible that a separate set of factors would emerge in contexts where, for example, tenure does not exist, or the tenure process and its criteria are different. Alternatively, investigating other contexts that employ different collaborative approaches could identify successful strategies for bringing ethical and scientific expertise together. In addition, it could be valuable to examine whether researchers in different areas of Human Genetics view the role and relevance of ELSI expertise differently. The genome scientists recruited for this study primarily focus on genetic and mechanistic discovery and methods development.

It is entirely feasible that genome scientists who more frequently interact with patients and study participants, like those in medical genetics, clinical genetics, and/or genetic epidemiology, might articulate different views. Finally, future research should investigate how ELSI scholars' and genome scientists' perceptions of the role and relevance of ELSI expertise change at different points in time, such as: over the course of a researcher's career, during the course of a specific collaboration, or in response to historical events.

Conclusion

The field of Human Genetics is finding itself having to navigate increasingly complex ethical and social issues (Green et al. 2020) while conducting research that is expansive in the audiences it works with and speaks to. In response, there have been calls for research collaborations between genome scientists and ELSI scholars (Martschenko and Trejo 2022; National Academies of Sciences, Engineering, and Medicine 2023; Reardon et al. 2023). These calls are grounded in the belief that such partnerships are central to ensuring genetic research is ethical and anticipatory in its prevention of harms and promotion of benefits. Nevertheless, there appear to be distinct differences between ELSI scholars and genome scientists in terms of how they perceive the role and relevance of ELSI expertise. Such differences may, in turn, make collaborations between ELSI scholars and genome scientists challenging to initiate and conduct. Thus, determining how to navigate the different values, norms, goals, conceptual frameworks, and research methods of genome scientists and ELSI researchers requires a deep understanding of the roles that these factors play in the development of the two research fields. Doing so could better support the research enterprise, and the genome scientists and ELSI scholars within it, to fulfill their social and ethical obligations to responsibly ask, answer, and communicate complex questions about humankind.

Notes

1. Originally we used only the keyword 'bioethics' to conduct our search for ELSI scholars. However, we soon realized that the use of this keyword resulted in a broader population than the one we intended to interview. As such, we refined our search terms to flag manuscripts that contained both the keywords 'ELSI' and 'bioethics.' We acknowledge that only using the keyword 'ELSI' would have sufficed.
2. After interview, one participant decided to remove themselves from the study due to concerns about being identifiable.

3. This potential participant declined to participate because they were retired and felt too removed from the field to be relevant.
4. The third-party transcription service involved a human for the entire process.
5. All three members of the research team conduct ELSI research – two of whom have faculty appointments at a bioethics center. We acknowledge our positionality as members of one of the two research cultures discussed in this paper. We feel it important to state our positionality and reflect on how it may shape a study exploring how ELSI scholars and genome scientists' think about their collaborations with each other. Furthermore, we recognize that for genome scientists', being interviewed by an ELSI scholar may have elucidated different responses than if they had been interviewed by a peer.
6. Participants were asked to provide information about their self-identified race, gender, and role by selecting from predetermined categories in a brief questionnaire that they received prior to interview. For race/ethnicity information, participants were able to select more than one category and/or to write-in their response. The survey items to collect gender and racial/ethnic self-identity were taken from the NIH All of Us Research program's "The Basics Survey." ([The Basics Survey – All of Us Research Hub](#)) The racial/ethnic survey identity was phrased as:
 - Which categories describe you? Note, you may select more than one group.
 - American Indian or Alaska Native (For example: Aztec, Blackfeet Tribe, Mayan, Navajo Nation, Native Village of Barrow Inupiat Traditional Government, Nome Eskimo Community, etc.).
 - Asian (For example: Asian Indian, Chinese, Filipino, Japanese, Korean, Vietnamese, etc.).
 - Black, African American, or African (For example: African American, Ethiopian, Haitian, Jamaican, Nigerian, Somali, etc.).
 - Hispanic, Latino, or Spanish (For example: Columbian, Cuban, Dominican, Mexican or Mexican American, Puerto Rican, Salvadoran, etc.).
 - Middle Eastern or North African (For example: Algerian, Egyptian, Iranian, Lebanese, Moroccan, Syrian, etc.).
 - Native Hawaiian or other Pacific Islander (For example: Chamorro, Fijian, Marshallese, Native Hawaiian, Tongan, etc.).
 - White (For example: English, European, French, German, Irish, Italian, Polish, etc.).
 - None of these fully describe me (if you want, describe a different category below in 'Other').
 - Other _____.

Participants were also asked open-ended questions about their training and current disciplines.

Authors' contributions

Conceptualization: D.O.M., M.K.C.; Investigation: D.O.M.; Formal Analysis: D.O.M., A.G.; Writing-original draft: D.O.M.; Writing-review & editing: M.K.C.; A.G.; Funding: M.K.C.

Data sharing

Redacted text transcripts that do not include any protected health information, individual names, addresses, specific geographic locations, and institution names will be made available to investigators upon reasonable request.

Disclosure statement

This manuscript is comprised of original material that is not under review elsewhere, and the subject on which the research is based has been subject to appropriate ethical review. The authors declare there are no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethics declaration

This study was approved by the Stanford IRB (protocol IRB-60608). Informed consent was required for this study and informed consent was obtained from all participants as required by the IRB.

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